



Soknarith (Soki) Sem

☎ (401) 952-6819 — ✉ seminaia401@gmail.com — 🌐 [Seminaia](#) — in [soknarith-sem](#) — 📍 Worcester, MA

SUMMARY

Materials and process engineer with experience in process optimization, manufacturing support, material characterization, and computational analysis across research and industrial environments. Background includes cGMP operations, statistical process control, root cause investigation, SOP development, and scale-up support. Brings a combined materials and process perspective to improving manufacturing efficiency, process reliability, and product quality.

WORK EXPERIENCE

Research Assistant / Graduate Teaching Assistant **May 2024 to Present**
Worcester Polytechnic Institute - Worcester, MA

- Quantified defect energetics and transport behavior in oxide perovskites using DFT and thermodynamic modeling to evaluate material stability and performance.
- Analyzed structure–property–process relationships across temperature and chemical potential regimes to understand defect behavior relevant to semiconductor manufacturing.
- Supported undergraduate and graduate materials engineering courses through grading and student instruction during homework help sessions.

Chemical Process Engineer **May 2023 to May 2024**
Knowles Precision Devices - New Bedford, MA

- Supported the R&D-to-production transition of high-voltage capacitor dielectric fluids, contributing to process scale-up and manufacturing readiness.
- Optimized manufacturing layout and process flow using Lean principles and statistical process control, reducing production costs by 15%.
- Authored SOPs and work instructions for material handling and process scale-up to support technician training and manufacturing consistency.
- Investigated material-related failures through root cause analysis, deviation support, and CAPA implementation to improve process reliability.

R&D Technician **Mar 2023 to May 2023**
Factorial Energy - Woburn, MA

- Fabricated Li-metal batteries in controlled glovebox and dry-room environments to support performance evaluation under varied operating conditions.
- Reduced material waste by 20% through redesign of the inventory system.

Downstream Purification Engineer **Jul 2022 to Mar 2023**
Eurofins Scientific - Framingham, MA

- Purified therapeutic antibodies and enzymes using AKTA Avant systems with Unicorn software, achieving 95–99% purity.
- Applied DOE methods to optimize buffer conditions and flow rate for improved yield and purity at lab scale.
- Performed membrane filtration and chromatography studies to support pilot-scale development and manufacturing scale-up.
- Characterized purified proteins using SDS-PAGE, HPLC/MS, GC/MS, UV-Vis, and related analytical techniques to confirm purity and identity.

EDUCATION

M.S. Materials Science & Engineering **GPA: 3.50**
Worcester Polytechnic Institute – Worcester, MA **2024 to 2026**

B.S. Chemical Engineering; B.S. Applied Mathematics **GPA: 3.16**
University of Rhode Island – Kingston, RI **2019 to 2022**

Minors: Chemistry, Physics

TECHNICAL SKILLS

Process & Manufacturing:	Process optimization, SPC, DOE, Lean/Six Sigma, SOP development, Tech Transfer (MSAT), cGMP
Analytical & Laboratory:	HPLC, AKTA FPLC, TFF, SDS-PAGE, UV-Vis, HPLC/MS, GC/MS, SEM, XRD, DSC/TGA, Osmometry, pH meter, KF titration, glovebox, clean-room
Scientific Computing:	Python, MATLAB, ASPEN Plus Dynamics, Linux, LaTeX, R, Unicorn
Simulation:	DFT (VASP & GPAW), Molecular Dynamics (LAMMPS)